Name:
Neptun code:

Midterm test
Quantum Physics
BME, physicist-engineer, 3th November

Only a calculator and pens may be used during the test. Mobile phones, tablets, laptops, etc. may not be used as calculators. The exercises must be solved individually. Start: 8:30 Submission: 10:00

You must write on the test paper with a blue pen. Write the final result in the designated box. Once a result has been written in the box, it may not be changed.

If needed, pages 3 and 4 may be used for detailed calculations, extra papers may be asked from the teacher.

All numerical solutions are accepted with a relative error up to 10%.

USEFUL FORMULAS

- reduced Planck constant: $\hbar = 1.034 \times 10^{-34}\,\mathrm{kg}\cdot\mathrm{m}^2/\mathrm{s}$.
- Quasi-degenerate perturbation theory (m: index from relevant subspace, l: index from irrelevant subspace)

$$\begin{aligned} S^{(1)}_{ml} &= -\frac{H'_{ml}}{E_m - E_l}, \\ H^{(0)}_{mm'} &= H^0_{mm'}, \\ H^{(1)}_{mm'} &= H'_{mm'}, \\ H^{(2)}_{mm'} &= \frac{1}{2}\sum_{l\in B} H'_{ml}H'_{lm'}\left[\frac{1}{E_m - E_l} + \frac{1}{E_{m'} - E_l}\right]. \end{aligned}$$

EXERCISES

1. *Characteristic Polynomial.* Compute the characteristic polynomial of

$$A = \begin{pmatrix} 3 & i \\ -i & 1 \end{pmatrix}$$

in the form $a\lambda^2+b\lambda+c$. What are the coefficients?

$$a = 1,\ b = -4,\ c = 2$$

2. *Eigenvalues of a Hermitian Matrix.* Find the eigenvalues of

$$H = \begin{pmatrix} 0 & 1-i \\ 1+i & 2 \end{pmatrix}.$$

$$\lambda_1 = 1+\sqrt{3},\ \lambda_2 = 1-\sqrt{3}$$

3. *Normalized Eigenvector.* Find the normalized eigenvector $\mathbf{v}$ corresponding to the eigenvalue $\lambda = 9$ of

$$H = \begin{pmatrix} 8 & -3i \\ 3i & 0 \end{pmatrix},$$

such that v_0 is real and positive.

$$v_0 = 3/\sqrt{10}, v_1 = i/\sqrt{10}$$

4. *Projector of a Vector.* Let $|v\rangle = \begin{pmatrix} 1 \\ i \end{pmatrix}$. Compute the projector $P = \frac{|v\rangle\langle v|}{\langle v|v\rangle}$.

$$P = \tfrac{1}{2}\begin{pmatrix} 1 & -i \\ i & 1 \end{pmatrix}$$

5. *Commutator of operators.* Given

$$A = \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix}, \quad B = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ 1 & 0 & 0 \end{pmatrix},$$

calculate: $[A, B] = AB - BA$.

$$[A,B] = \begin{pmatrix} 0 & -1 & 0 \\ 1 & 0 & 1 \\ 0 & -1 & 0 \end{pmatrix}$$

6. *Pauli Decomposition.* Write

$$H = \begin{pmatrix} 2 & 1-i \\ 1+i & 0 \end{pmatrix}$$

as $H = a_0 I + \boldsymbol{a}\cdot\boldsymbol{\sigma}$, where a_0 and the components of $\boldsymbol{a}$ are real.

$$a_0 = 1 \quad a_x = 1$$

$$a_y = 1 \quad a_z = 1$$

7. *Time-evolution operator.* Given

$$H = \frac{\hbar\omega}{2\sqrt{2}}(\sigma_x + \sigma_z),$$

compute

$$U(t) = e^{-iHt/\hbar} = a_0 I + i\boldsymbol{a}(t)\cdot\boldsymbol{\sigma},$$

where a_0 and the components of $\boldsymbol{a}(t)$ are real.

$$a_0 = \cos(\omega t/2) \quad a_x = -\sin(\omega t/2)/\sqrt{2}$$

$$a_y = 0 \qquad a_z = -\sin(\omega t/2)/\sqrt{2}$$

8. *Expectation Values.* For $|+\rangle = \frac{1}{2}(|0\rangle + \sqrt{3}\,|1\rangle)$, compute $\langle\sigma_x\rangle$.

$$\langle\sigma_x\rangle = \sqrt{3}/2$$

9. *Larmor precession.* Given $H = \frac{\hbar\omega}{2}\sigma_y$ and initial state $|\psi_0\rangle = (|0\rangle + |1\rangle)/\sqrt{2}$, calculate the probability of measuring the $+1$ eigenvalue of the observable σ_z after time t. *Hint:* $\cos x + \sin x = \sqrt{2}\sin(x + \pi/4)$.

$$p_{+1}(t) = \sin(\omega t/2 + \pi/4)$$

10. *Time-dependent unitary transformation.* Calculate the Hamiltonian $H = \frac{\hbar\omega}{2}\sigma_z + \hbar\Omega\sigma_x\cos(\omega t)$ in the rotating frame where the static term $\frac{\hbar\omega}{2}\sigma_z$ is eliminated. Expand as a linear combination of Pauli matrices:

$$\tilde{H}(t) = a_0(t)I + \boldsymbol{a}(t)\cdot\boldsymbol{\sigma}.$$

$$a_0(t) = 0 \qquad a_x(t) = \cos^2(\omega t)$$

$$a_y(t) = -\cos(\omega t)\sin(\omega t) \qquad a_z(t) = 0$$

11. *Ladder operators.* Calculate the matrix element $\langle n|\, a^\dagger aaa^\dagger\, |n\rangle$, where a and $a^\dagger$ are the ladder operators of the harmonic oscillator, and $|n\rangle$ is the eigenstate of the harmonic oscillator with energy $\hbar\omega(n + 1/2)$.

$$\langle n|\, a^\dagger aaa^\dagger\, |n\rangle = n(n+1)$$

12. *Expectation value of x in superposition.* Express the expectation value of x with the oscillator length ℓ, in the state $|\psi\rangle = \frac{1}{2}(\sqrt{3}\,|0\rangle + |1\rangle)$, where $|0\rangle$ and $|1\rangle$ are the eigenstates of the harmonic oscillator.

$$\langle x\rangle = \sqrt{3/8}\ell$$

13. *Mass-and-spring.* For a mass-and-spring system with $m = 9.1\cdot 10^{-31}$ kg and $k = 10^{-16}$ N/m, calculate the oscillator length.

$$\ell = 3.24\ \mu\text{m}$$

14. *Non-degenerate perturbation theory.* Consider

$$H = \frac{\hbar\omega_0}{2}\sigma_x + \frac{\hbar\omega_1}{2}\sigma_y,$$

with $\omega_1 \ll \omega_0$.

Calculate the energy of the ground state up to second order in $\nu = \omega_1/\omega_0$ in the form:

$$E_0 = -\frac{\hbar\omega_0}{2}(1 + a\nu^2)$$

$$a = 1/2$$

15. *Quasi-degenerate perturbation theory: effective Hamiltonian.* Consider

$$H = H^0 + H',$$

$$H^0 = \begin{pmatrix}\Delta & 0 & 0\\ 0 & 0 & 0\\ 0 & 0 & 0\end{pmatrix}, \quad H' = \begin{pmatrix}0 & t & 0\\ t & 0 & t\\ 0 & t & 0\end{pmatrix},$$

with $\Delta \gg t > 0$ real parameters. Using quasi-degenerate perturbation theory, calculate the effective Hamiltonian corresponding to the doubly degenerate ground-state up to second order.

$$H_{\text{eff}} = \begin{pmatrix}-t^2/\Delta & t\\ t & 0\end{pmatrix}$$

16. *Quasi-degenerate perturbation theory: subspace correction.* Consider

$$H = H^0 + H',$$

$$H^0 = \begin{pmatrix}\Delta & 0 & 0\\ 0 & 0 & 0\\ 0 & 0 & 0\end{pmatrix}, \quad H' = \begin{pmatrix}0 & t & 0\\ t & 0 & t\\ 0 & t & 0\end{pmatrix},$$

with $\Delta \gg t > 0$ real parameters. Calculate S up to first order, for which $\tilde{H} = e^{-S}He^S$ is block diagonal, i.e., $\tilde{H}_{01} = \tilde{H}_{02} = 0$.

$$S = \begin{pmatrix}0 & -t/\Delta & 0\\ t/\Delta & 0 & 0\\ 0 & 0 & 0\end{pmatrix}$$

17. *States of composite systems.* Given

$$|\psi\rangle = \frac{1}{\sqrt{5}}\begin{pmatrix}i\\ 2\end{pmatrix}, \quad |\phi\rangle = \frac{1}{2}\begin{pmatrix}\sqrt{2}\\ 1\\ 1\end{pmatrix},$$

calculate $|\psi\rangle \otimes |\phi\rangle$.

$$|\psi\rangle \otimes |\phi\rangle = 1/2\sqrt{5}\begin{pmatrix}i\sqrt{2}\\ i\\ i\\ 2\sqrt{2}\\ 2\\ 2\end{pmatrix}$$

18. *Operators of composite systems.* Calculate $\sigma_y \otimes \sigma_z$.

$$\sigma_y \otimes \sigma_z = \begin{pmatrix} 0 & 0 & -i & 0 \\ 0 & 0 & 0 & i \\ i & 0 & 0 & 0 \\ 0 & -i & 0 & 0 \end{pmatrix}$$

19. *Eigenvalues of an operator of a composite system.* Calculate the eigenvalues of $\sigma_y \otimes \sigma_y$.

$$\lambda_1 = -1 \quad \lambda_2 = -1$$

$$\lambda_3 = 1 \quad \lambda_4 = 1$$

20. *Subsystem observables.* Given the composite state of two-qubits

$$|\psi\rangle = \frac{1}{3} \begin{pmatrix} 2 \\ 1 \\ i \\ \sqrt{3} \end{pmatrix},$$

calculate the probability, of measuring the $+1$ eigenvalue of the local observable $\sigma_x \otimes I$.

$$p_{+1} = (9 + 2\sqrt{3})/18$$

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